Coating-Inspection — Standard Operating Procedure

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Coating-Inspection — Standard Operating Procedure

> **Audience:** Quality Worker · Maintenance Tech · Line Manager
> **Applies to:** Line-D, station 6, Coating-Inspection

1. Purpose

Coating-Inspection verifies finish quality (thickness, color, surface defects) on every cured part before Final-Packaging. Reject disposition follows `Quality_Reject_Disposition_Policy.pdf`.

2. Normal operating envelope

- Ideal cycle time: **15 s**
- Max acceptable cycle time: **20 s**
- Expected reject rate: **≤ 3 %**
- Fault probability per cycle: **0.2 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

`idle_reason`	Meaning here	First check
`Starved`	Curing-Oven (station 5) is not feeding parts.	Verify Curing-Oven state — the line bottleneck.
`Blocked`	Final-Packaging (station 7) cannot accept the part.	Verify Final-Packaging state.

4. Faults & corrective actions

4.1 `Camera-Fault`

- **Symptoms in telemetry:** `fault_type = "Camera-Fault"`. Inspection cycle aborts or rejects spike with vision-system codes.
- **Likely root cause:** lens fouled, focus drifted, or camera controller failure.
- **Corrective action:**

1. Clean lens; check focus.
2. Cycle camera power.
3. Replace camera if image acquisition still fails.

- **Restart criteria:** master part passes vision recipe.

4.2 `Light-Fail`

- **Symptoms in telemetry:** `fault_type = "Light-Fail"`. Image contrast drops; rejects surge.
- **Likely root cause:** light bar driver failed or LED bank ended life.
- **Corrective action:**
  1. LOTO. Replace failed light.
  2. Restore brightness setpoint.
- **Restart criteria:** image meets contrast threshold.

4.3 `Sensor-Drift`

- **Symptoms in telemetry:** `fault_type = "Sensor-Drift"`. False-positive rejects rise; thickness gauge readings vary.
- **Likely root cause:** thickness gauge calibration drift or temperature effects.
- **Corrective action:**
  1. Run calibration routine against master step block.
  2. Replace sensor if out of spec.
- **Restart criteria:** calibration error within ± 2 µm.

5. Preventive maintenance schedule

- **Daily:** wipe lenses; run master parts at shift start.
- **Weekly:** clean diffusers.
- **Monthly:** thickness gauge calibration.
- **Quarterly:** lighting replacement.

6. References

- `Quality_Reject_Disposition_Policy.pdf`
- `Maintenance_Workflow_Overview.pdf`
- `Shift_Handover_Checklist.pdf`
- `Line-D_05_Curing-Oven_SOP.pdf`
- `Line-D_07_Final-Packaging_SOP.pdf`